OBJECT ORIENTED AND FUNCTIONAL PROGRAMMING WITH PYHTON

TASK 1: Creating a Habit Tracking App

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CONCEPTION

Habits are basically small tasks or actions one does regularly with little or no effort and is part of their lifestyle. This project is aimed at building a basic backend CLI based habit tracking app to help users create habits, keep track of habits and achieve personal goals. The user should be able to add multiple tasks, analyze tasks, check off tasks, keep streaks and remove tasks.

To support these functionalities, the system is composed of 5 core components that work together to manage habit data, user interactions and analytic operations.

a) **Habit Class:** Habit class represents each individual habit with all the data and behavior associated with it. It adopts an objected oriented approach to ensure clear and structured representation of habits data like habit name, description, completion history and time created. It makes the habit class reuseable, maintainable and ensures data encapsulation.

b) **Habit Manager:** The habit manager acts as the central controller, managing all the habits in the app. It ensures the habit is created, loaded, uploaded, deleted and saved consistently to the JSON file. The manager handles the storage and retrieval allowing the rest of the application to focus on its intended functionalities.

c) **Analytics Module:** The analytics modules is implemented using a functional approach. It focuses on processing input data and producing clear outputs without altering existing data. It is used for tasks like calculating habit streaks, grouping habits periodically and generating insights of the user activity. This makes the system easier to understand, test and maintain.

d) **JSON Files (Persistence layer):** The application stores data in a JSON file in a structured and readable manner. When the user interacts with the application like completing a habit, the changes are immediately saved. It is simple, light weighted, human readable and flexible.

e) **User Interface:** The application is operated through Command Line Interface. When the application is launched, the user is presented with a menu of options such as create habit, list of habits, marking habits as completed and viewing streaks. The display text is made colorful and some tabular to enhance display on the console and more readable. This approach keeps the interface simple and ensures clear separation between user interface and application logic.

In summary, applies the principle of Object Oriented Programming (OOP) to model the habits and organize them and Functional Programming to perform analytical task producing clean and reliable outputs. JSON offers a simple method for data persistence and the CLI allows users interact with the application. All these ensures the system fulfills the project's functional and technical requirements.

The work flow of the application will look like